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How To Backup Ubuntu And Restore It Using "Dd"



DD is a Unix and Unix-like utility that is primarily used to convert and copy files. One really useful use-case for dd is that it can "[pack*](#)" entire partitions into one single image file, thus allowing users to recover entire disks by "[unpacking*](#)" and restoring a file-system in case of hardware failure.

To create a full backup for a partition the command you need to run is straight-forward. The size of the resulting image file will be the same as the size of your partition so make sure you have enough free space on the drive you are placing your backup... also, please avoid placing your image file on the same drive that you're backing up.

FIRST, LET'S IDENTIFY WHICH PARTITION WE SHOULD BACKUP: `USER@X:~$ df`

The output of `df` (above) will look similar to this:

```
codepunker@codepunker-P5K-SE: ~  
codepunker@codepunker-P5K-SE:~$ df  
Filesystem      1K-blocks      Used Available Use% Mounted on  
/dev/sda1       107862000  55834684  46525120  55% /  
none            4          0          4    0% /sys/fs/cgroup  
udev            8047528     4      8047524    1% /dev  
tmpfs           1611664     1468    1610196    1% /run  
none            5120        0       5120     0% /run/lock  
none            8058316     58520    7999796    1% /run/shm  
none            102400      72     102328    1% /run/user  
/dev/sdb1       976759804  597751560  379008244  62% /mnt/CEE6C1E9E6C1D1BF  
/dev/sdc1       154090492  26285240  127805252  18% /media/codepunker/bkp2  
/dev/sdc2       471037948  278325508  192712440  60% /media/codepunker/bkp1  
codepunker@codepunker-P5K-SE:~$ _
```

You will know that which partition is your primary one by looking at the "Mounted on" column... and identifying the entry that is "Mounted on" "/".

NOW YOU'RE READY TO BACKUP YOUR FILE-SYSTEM

```
user@x:~$ sudo dd if=/dev/sda1 of=/backup/location/filename.img
```

The backup process will take a while so sit tight and let it to its job. Once it finishes it will show you a report with X records in, X records out, X seconds etc....

SETTING THIS UP AS A CRON-JOB

You probably don't want to run this manually simply because running backups should be a scheduled process, so I suggest you set this up as a weekly or even daily cron job. To enable this run: `sudo crontab -e` and add the line below making sure you modify the paths and time to fit your needs.

```
@weekly dd if=/dev/sda1 of=/backup/location/filename.img >> /var/log/backup.log
```

RESTORING

In case you need to restore the image you've just created ... which I hope never happens :) ... all you need to do is to boot your system using a live Ubuntu cd or bootable usb stick and run the dd command in reverse pointing to a newly created partition, on a new hard-drive.

```
user@x:~$ sudo dd if=/backup/location/filename.img of=/dev/sda1
```

I ALSO MADE A SHORT VIDEO SHOWING THE ABOVE

Backup Ubuntu With One Terminal Command



* Really... dd is not an archive manager, but I didn't find a more suitable word for what actually happens.

** You need to run `sudo crontab -e` because you need root privileges